

ECEn 471 — Homework 6

Please work the following problems. There is a small dataset at the bottom.

1. In this problem you will plot a number of loss functions and compare them.
 - (a) Plot the loss functions from the text for regression to reproduce Figure 5.1. List the loss functions from the most robust to the least robust.
 - (b) Plot the loss functions from the text for classification to reproduce Figure 5.2. List the loss functions from the most robust to the least robust.
2. In Problems 2–5, you will explore how regularization affects model complexity and generalization performance.

Suppose we observe data generated from the function

$$y = \sin(2\pi x) + \epsilon$$

where $\epsilon \sim \mathcal{N}(0, 0.1^2)$ is Gaussian noise.

- (a) Generate 25 training points (x_i, y_i) where $x_i \sim \text{Uniform}[0, 1]$, meaning the continuous uniform distribution ranging from zero to one.
 - (b) Generate 200 test points using the same distribution.
 - (c) Plot the training points and the test points on the same plot using different colors, and yes, you read the numbers correctly. We have more test data than training data on purpose to ensure that the problem will explore the potential to overfit the model to the training data.
3. Fit a polynomial regression model of degree 10 to the data you generated in Problem 2. That is, find θ to minimize the squared-error loss assuming the model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_{10} x^{10}.$$

- (a) Construct the matrix $\mathbf{X}$ for the polynomial model in your code.
- (b) Compute the least-squares estimate

$$\hat{\theta} = \arg \min_{\theta} \|y - \mathbf{X}\theta\|_2^2.$$

- (c) Plot the fitted polynomial curve together with the training data.
- (d) Compute the mean-squared error on both the training set and the test set.

4. Now consider the regularized optimization problem for the data you generated in Problem 2:

$$\hat{\theta} = \arg \min_{\theta} \|y - \mathbf{X}\theta\|_2^2 + \lambda \|\theta\|_2^2.$$

- (a) Solve this problem for the following values of λ :

$$\lambda \in \{0, 10^{-4}, 10^{-2}, 10^{-1}, 1\}.$$

Note that the $\lambda = 0$ case is the case with no regularization from the previous problem. You may also want to consult Equation (5.24) from your textbook for this part.

- (b) For each value of λ :
 - i. Plot the fitted polynomial curve.
 - ii. Compute the training and test MSE.
 - (c) Plot training error and test error as a function of λ .
5. Think a bit more deeply about Problem 4 to answer the following questions.
- (a) What happens to the fitted curve when $\lambda = 0$?
 - (b) Why does increasing λ initially improve test error?
 - (c) What happens when λ becomes too large?
 - (d) Based on your results, explain how regularization affects overfitting.
6. Consider the “Time to reflect 5.2” table on page 114 in the textbook. Fill out the table for only the first six rows. For the sake of clarity, the table is properly filled out with check marks indicating where the tools apply. For each row, write 1-2 sentences explaining your table entries.
7. Consider the gradient descent technique for the binary logistic regression cost function, where

$$J(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \ln(1 + \exp(-y_i \boldsymbol{\theta}^T \mathbf{x}_i)).$$

Prove that equation (5.29) from your textbook is correct.

8. Consider the Training Dataset below. Suppose you want to use logistic regression with the cross-entropy loss function to predict the author of General Conference talks using the features presented (the percentages of words in each talk equal to “the” and “of”). For our toy example here, assume that all talks are either authored by Elder Bednar or President Nelson. Do not use the built-in functions from Sci-kit learn, although you may use them to check your work. Use gradient descent with the answer from the previous question. Pick a reasonable step size and stopping criterion for your optimizer.
- (a) Using only the percentage of words equal to “the”, form a logistic regression model. Give the form and all parameters for your model. Justify your choice of γ and your stopping criterion.
 - (b) Repeat part (a) using both features.
 - (c) Suppose that deep in the archives of the Church of Jesus Christ of Latter-day Saints, a General Conference talk is found with features $\mathbf{x}_* = [6.86, 4.75]$. At the top of the talk, someone has written the sentence, “This talk was either written by Elder Bednar or President Nelson, but are you clever enough to determine which of us wrote it?” Undaunted, you apply both of your logistic regression models from this problem. Who wrote the talk, according to your models? What are the probabilities $\Pr(\text{Bednar}|\mathbf{x}_*)$ and $\Pr(\text{Nelson}|\mathbf{x}_*)$ in both cases?
 - (d) Which of your models do you think is better, i.e., do you think the % words = “of” stat improves your model?

Table 1: Training Dataset — General Conference Talks (from churchofjesuschrist.org)

Talk Name	$y = \text{Author}$	$\mathbf{x}^T = [\% \text{ words} = \text{"the"}, \% \text{ words} = \text{"of"}]$
Let This House Be Built unto My Name	Elder Bednar	[10.43, 5.72]
We Will Prove Them Herewith	Elder Bednar	[5.71, 4.14]
With the Power of God in Great Glory	Elder Bednar	[9.18, 6.18]
But We Heeded Them Not	Elder Bednar	[8.82, 5.15]
Embrace the Future with Faith	President Nelson	[5.23, 2.47]
Christ is Risen; Faith in Him Will Move Mountains	President Nelson	[5.66, 3.14]
The Temple and Your Spiritual Foundation	President Nelson	[6.74, 2.73]
The Power of Spiritual Momentum	President Nelson	[5.28, 2.91]